Digital Chip Design Program

RTL Design

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Task #5

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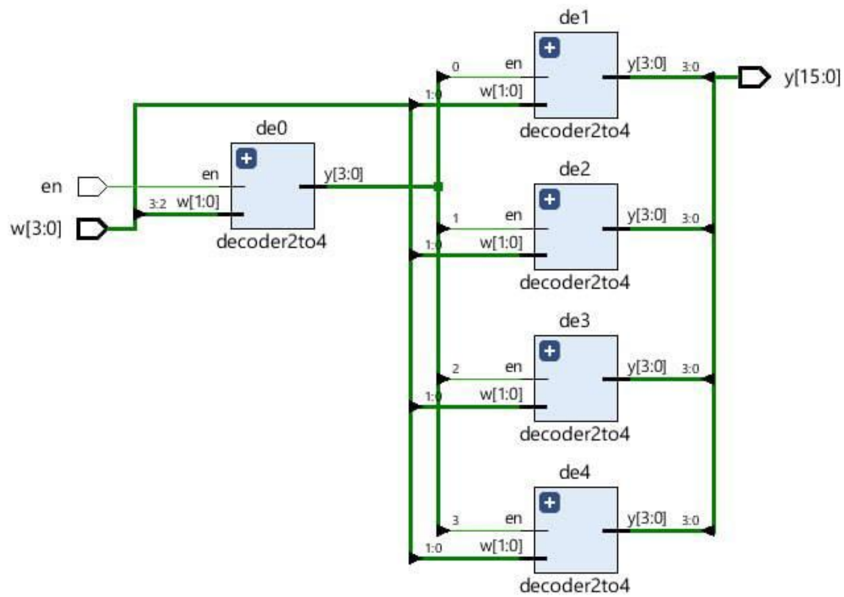
Combinational Datapath Design Part 1

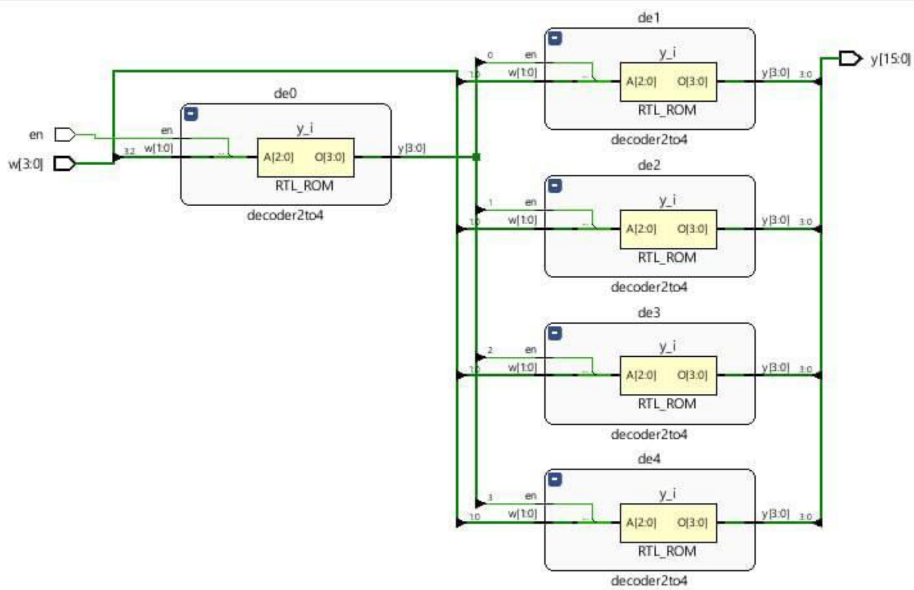
Task #1.1 (1.1.1)

```
module decoder2to4(
input logic [1:0] w,
input logic en,
output logic [3:0]y
);
always @ (w or en)
case({w,en})
3'b100 : y=4'b1000;
3'b101 : y=4'b0100;
3'b110 : y=4'b0010;
3'b111 : y=4'b0001;
default : y=4'b0000;
endcase

endmodule
```

(1.1.2)

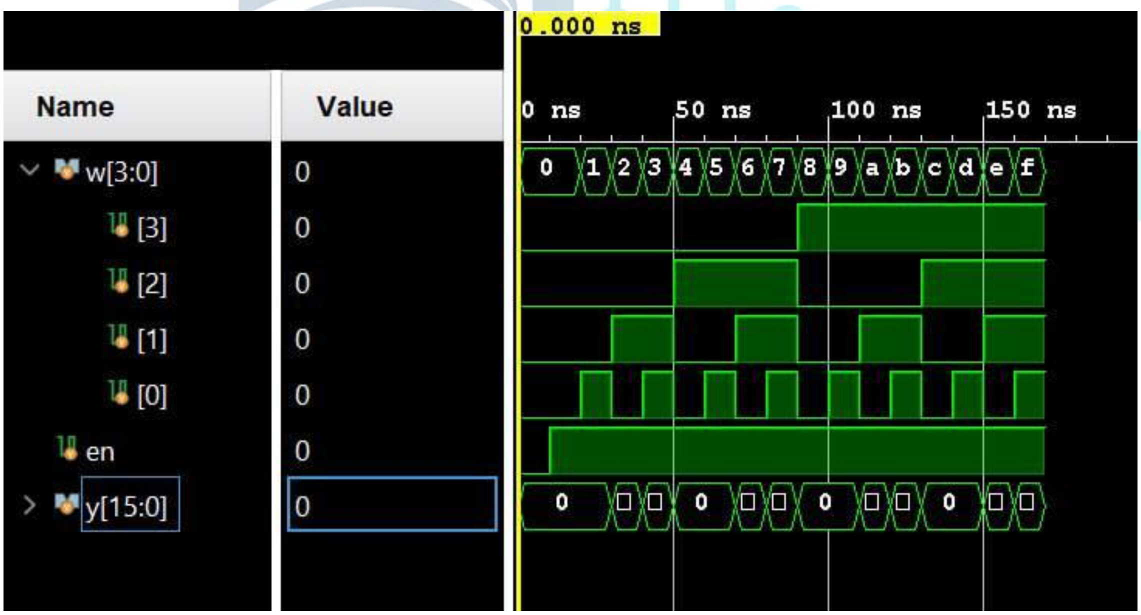
```
module decoder4to16(
input logic [3:0] w ,
input logic en ,
output logic [15:0]y

);
logic [3:0]m ;

decoder2to4 de0(.w(w[3:2]) , .en(en) ,.y(m));
decoder2to4 de1(.w(w[1:0]) , .en(m[0]),.y(y[3:0] ));
decoder2to4 de2(.w(w[1:0] ), .en(m[1]),.y(y[7:4]));
decoder2to4 de3(.w(w[1:0]) , .en(m[2]),.y(y[11:8]));
decoder2to4 de4(.w(w[1:0]) , .en(m[3]),.y(y[15:12]));


endmodule
```


de1
en
w[1:0]
y[3:0]
decoder2to4
y[15:0]
de0
en
w[1:0]
y[3:0]
decoder2to4
en
w[3:0]
de2
decoder2to4
de3
decoder2to4
de4
decoder2to4


de1
y_i
A[2:0]
O[3:0]
RTL_ROM
decoder2to4
y[15:0]
de0
decoder2to4
en
w[3:0]
de2
decoder2to4
de3
decoder2to4
de4
decoder2to4

(1.1.3)

```
`timescale 1ns / 1ps

module testbench();
logic [3:0] w ;
logic en ;
logic [15:0]y ;

decoder4to16 dut(
.w(w),
.en(en),
.y(y)
);

initial begin
en=0;
w=4'b0000;
#10;
en=1;
for(int i=0; i<16;i++)begin
w=i;
#10;
end
$finish;
```



Task #1.2(1.2.1)

```
module HalfAdder (
input logic A,B,
output logic s, c
) ;

assign {c, s} = A + B;
endmodule
```

(1.2.2)

```
module FullAdder (
input logic A, B, cin,
output logic cout, s
);
logic c1, s1, c2;

HalfAdder HA1 (A, B, s1, c1);
HalfAdder HA2 (s1, cin, s, c2);

assign cout = c1 | c2;

endmodule
```

(1.2.3)

```
module RippleAdder(
input logic [3:0] A, B,
input logic cin,
output logic [3:0] s,
output logic cout
);

logic c1, c2, c3;

FullAdder F1 (A[0], B[0], cin, c1, s[0]);
FullAdder F2 (A[1], B[1], c1, c2, s[1]);
FullAdder F3 (A[2], B[2], c2, c3, s[2]);
FullAdder F4 (A[3], B[3], c3, cout, s[3]);

endmodule
```

(1.2.4)

```
module TopCircuit (
input logic [3:0] A, B,
input logic M,
output logic [3:0] S,
output logic Cout
);

logic [3:0] sub_B;



assign sub_B = (M == 1'b1) ? ~B : B;

RippleAdder adder_sub (
.A(A),
.B(sub_B),
.cin(M),
.s(S),
.cout(Cout)
);

endmodule
```

(1.2.5)

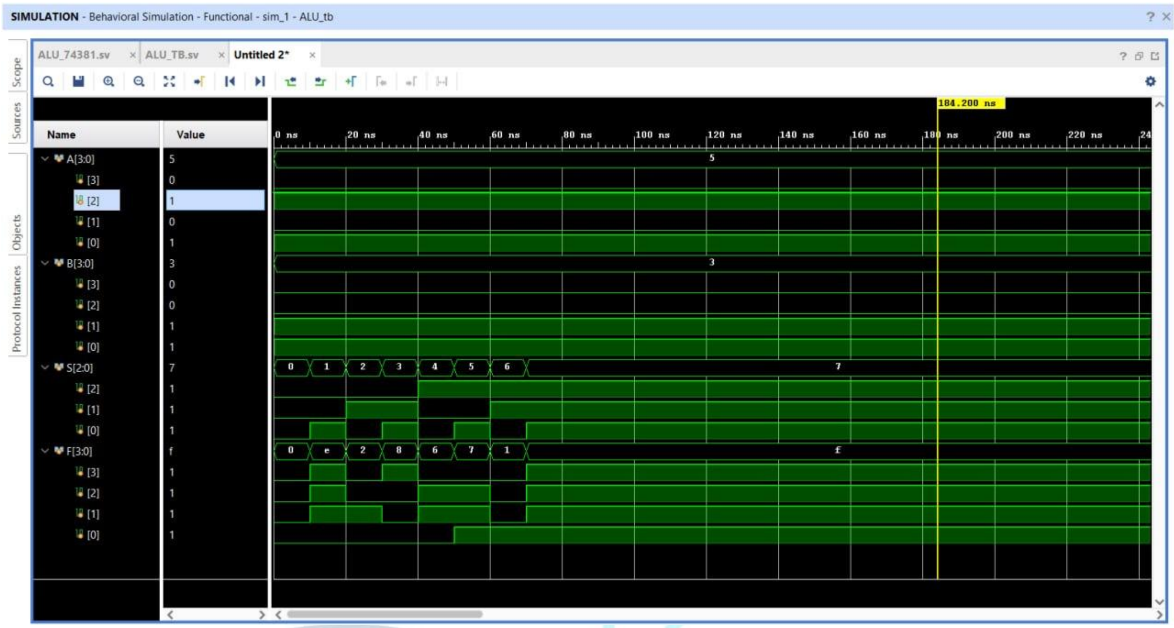
```
`timescale 1ns / 1ps
module TopCircuit_tb;

logic [3:0] A, B;
logic M;
logic [3:0] S;
logic Cout;

integer i;

TopCircuit uut (
.A(A),
.B(B),
.M(M),
.S(S),
.Cout(Cout)
);

initial begin
M = 0;
for (i = 0; i < 10; i = i + 1) begin
A = i;
B = i + 1;
#10;
end

M = 1;
for (i = 0; i < 10; i = i + 1) begin
A = i + 5;
B = i;
#10;
end

$finish;
end

endmodule
```

Task #1.3

```
module ALU_74381(
input logic [3:0] A,
input logic [3:0] B,
input logic [2:0] S,
output logic [3:0] F
);

always_comb begin
case (S)
3'b000: F = 4'b0000;
3'b001: F = B - A;
3'b010: F = A - B;
3'b011: F = A + B;
3'b100: F = A ^ B;
3'b101: F = A | B;
3'b110: F = A & B;
3'b111: F = 4'b1111;
default: F = 4'b0000;
endcase
end
endmodule
```

Task #1.3

```
module ALU_tb;

logic [3:0] A;
logic [3:0] B;
logic [2:0] S;
logic [3:0] F;

ALU_74381 uut (
.A(A),
.B(B),
.S(S),
.F(F)
);

initial begin
A = 4'b0101;
B = 4'b0011;

S = 3'b000; #10;
S = 3'b001; #10;
S = 3'b010; #10;
S = 3'b011; #10;
S = 3'b100; #10;
S = 3'b101; #10;
S = 3'b110; #10;
S = 3'b111; #10;

$finish;
end

endmodule
```


SIMULATION - Behavioral Simulation - Functional - sim_1 - ALU_tb
ALU_74381.sv
ALU_TB.sv
Untitled 2*
Name
Value
A[3:0] 5
[3] 0
[2] 1
[1] 0
[0] 1
B[3:0] 3
[3] 0
[2] 0
[1] 1
[0] 1
S[2:0] 7
[2] 1
[1] 1
[0] 1
F[3:0] f
[3] 1
[2] 1
[1] 1
[0] 1
184.200 ns
0 ns
20 ns
40 ns
60 ns
80 ns
100 ns
120 ns
140 ns
160 ns
200 ns
220 ns
0 1 2 3 4 5 6 7
0 e 2 8 6 7 1 f

Work distribution

Section	ROAA	RAND	LANA	LARA	RAWAN	RENAD
Report					✓	✓
Task 1.1					✓	✓
Task 1.2			✓	✓		
Task 1.3	✓					